

Downstream Applications

3D Reconstruction

Back-project depth to point cloud:

$$X = \frac{(u - u_0) \cdot D}{f}$$

$$Y = \frac{(v - v_0) \cdot D}{f}$$

$$Z = D$$

Table 6: competitive with MVS methods using a *single* image.

SLAM Improvement

Using Metric3D v2 as depth prior in visual SLAM:

- ▶ 10× reduction in drift (Table 7)
- ▶ Enables metric-scale monocular SLAM

Our Point Cloud Demo

Point cloud rendering
(replace with
PLY render)